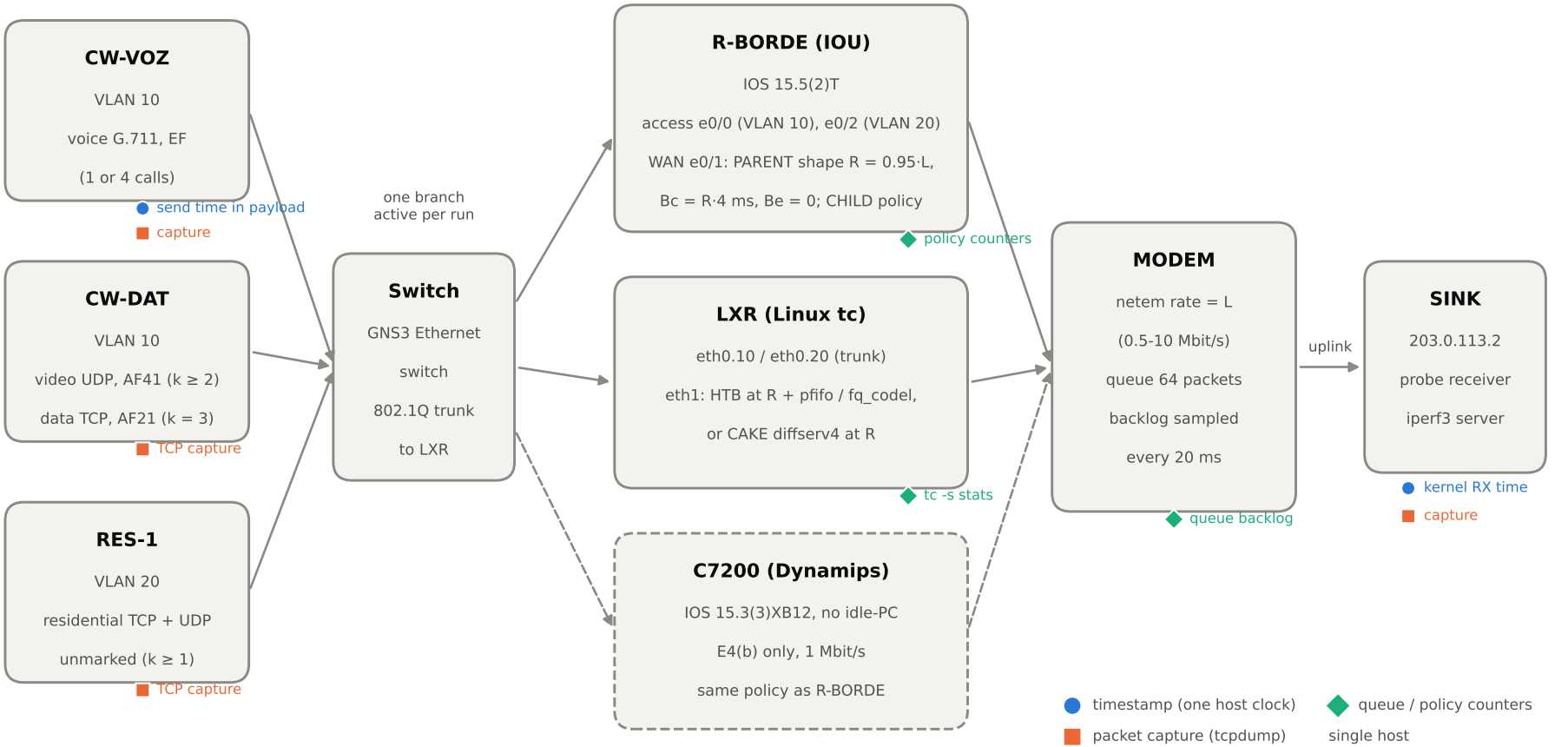
CW-VOZ
VLAN 10
voice G.711, EF
(1 or 4 calls)
send time in payload
capture
CW-DAT
VLAN 10
video UDP, AF41 (k ≥ 2)
data TCP, AF21 (k = 3)
TCP capture
RES-1
VLAN 20
residential TCP + UDP
unmarked (k ≥ 1)
TCP capture
one branch
active per run
Switch
GNS3 Ethernet
switch
802.1Q trunk
to LXR
R-BORDE (IOU)
IOS 15.5(2)T
access e0/0 (VLAN 10), e0/2 (VLAN 20)
WAN e0/1: PARENT shape R = 0.95·L,
Bc = R·4 ms, Be = 0; CHILD policy
policy counters
LXR (Linux tc)
eth0.10 / eth0.20 (trunk)
eth1: HTB at R + pfifo / fq_codel,
or CAKE diffserv4 at R
tc -s stats
C7200 (Dynamips)
IOS 15.3(3)XB12, no idle-PC
E4(b) only, 1 Mbit/s
same policy as R-BORDE
MODEM
netem rate = L
(0.5-10 Mbit/s)
queue 64 packets
backlog sampled
every 20 ms
queue backlog
uplink
SINK
203.0.113.2
probe receiver
iperf3 server
kernel RX time
capture
timestamp (one host clock)
packet capture (tcpdump)
queue / policy counters
single host